

Prior Art Search & Patentability Analysis: On-Device LLM → Closed-Graph Navigation as a Native Nav2 BehaviorTree Plugin

TL;DR

- **No single prior-art reference anticipates the full claimed combination**, but four of your five individual "novelties" are wholly or partly anticipated by existing work; only the tight *combination* of (closed-set annotation-graph constraint used as an explicit safety boundary) + (LLM inference embedded inside a Nav2 BehaviorTree plugin via pluginlib) + (deployment-log-trained confidence calibration gate) is defensible.
- The strongest blocking/adjacent references are **Google's US20230311335A1 / WO2023192497A1 (the "Natural language control of a robot" / SayCan patent family)**, **KnowNo (arXiv 2307.01928, conformal calibration for LLM planners)**, **SPINE (arXiv 2410.03035, validated graph goals as a safety mechanism)**, **osmAG-LLM (arXiv 2507.12753, LLM restricted to named nodes)**, and **Das, Hussain & Nawaz 2026 (DOI 10.3390/s26020608, LLM+Nav2 benchmarking)** — none combines all of your elements.
- **Recommended action:** file an Indian provisional immediately claiming the *combination*, drafted around concrete "technical effects" (physical safety from a hallucination-proof goal set + bounded in-tick latency) to survive Section 3(k), then a PCT within 12 months.

Key Findings

1. **On-device LLM for navigation is NOT novel on its own.** Multiple 2025–2026 works run a language model on the robot's embedded compute with no cloud: the "Plug-and-Play LLM Knowledge Extraction" framework (DOI 10.3390/make8020049) runs DistilGPT-2 fully on-device on a Jetson AGX Orin; LocalNav (arXiv 2606.27871) distills a 4B VLM for on-device object-goal navigation on a Jetson Orin; Qwen-RobotNav runs on-device on a Jetson Thor.
2. **Constraining an LLM to a closed set of options is well-established.** SayCan constrains an LLM to a fixed repertoire — its skill repertoire is **551 primitive actions across seven families (picking, placing, moving, knocking, upright, drawer manipulation, and navigation)**, and PaLM-SayCan "chooses the correct sequence of skills 84% of the time and executes them successfully 74% of the time" (Ahn et al. 2022, arXiv 2204.01691). But framing a *closed set of named location nodes* as a *safety boundary against hallucination* has no exact match in the searched patents or papers.
3. **LLMs already interoperate with Nav2 and BehaviorTrees**, but as external servers/generators — not as the inference engine embedded inside a pluginlib BT action node. **BTGenBot (Izzo, Bardaro & Matteucci, IROS 2024, arXiv 2403.12761)** fine-tunes Llama2-7b, Llama-Chat and CodeLlama-7b via LoRA on ~600 behavior trees to *generate BT XML*; its "MoveTo" node "searches [the location name] in a YAML file that maps location names to their respective coordinates," with the client + LLM running on an NVIDIA Jetson AGX Orin. The LLM is not itself a BT plugin.
4. **Calibrated confidence gating for LLM robot planners is anticipated in principle by KnowNo** (Ren et al., Princeton/Google DeepMind, CoRL 2023, arXiv 2307.01928): "if the prediction set at a scenario is not a singleton... KnowNo deems the robot uncertain... and triggers human help," explicitly addressing LLMs "prone to confidently hallucinated predictions." This weakens claim 4 substantially.
5. **Runtime-updatable maps/parameters via ROS services are routine engineering** (*dynamic_reconfigure* in ROS 1; built-in reconfigurable parameters in ROS 2), so claim 5 is anticipated and should be a dependent claim only.

Details

Prior Art Matrix

ID	Title / Venue	Year	What it discloses	Anticipates which of the 5 claims
US20230311335A1 / WO2023192497A1	"Natural language control of a robot" (Google LLC; inventors Hausman, Ichter, Levine, Toshev, Xia, Parada)	2023	LLM processes free-form NL instruction; a task-grounding measure (skill-description probability in the LLM output) is combined with a world-grounding (affordance) measure to select a robotic skill, including "navigate to X"; skill set is enumerated.	C1 partial (LLM→nav goal); C2 partial (enumerated skill set); C4 partial (probability weighting, not calibrated).
US20210086353A1 / US11972339B2	"Controlling a robot based on free-form natural language input" (Google)	2021 / 2024	Deep-RL model maps free-form NL + sensor data to actions; NL directs robot to landmarks/waypoints; reward on reaching a waypoint.	C1 partial (NL→waypoint); no LLM, no closed graph.
US11151992B2	"Context aware interactive robot" (semantic location map, voice)	2021	Robot builds a semantic location map, guides a customer to a named product location via voice, with autonomous obstacle avoidance.	C2 partial (named locations); C5 partial (map building). No LLM.
US9758305B2 / US10019015	"Robotic navigation utilizing semantic mapping" (Locus Robotics)	2017 / 2018	Warehouse robot maps SKU→fiducial ID→coordinates; navigates to an enumerated named fiducial. Classified G05D1/02.	C2 partial (closed set of named/fiducial goals). No language model.
CN106840141A	"Semantic map of mobile robot indoor navigation"	2017	Predicate-logic semantic map of named indoor locations for navigation/path planning.	C2 partial (named-location closed set).
US11813751 / US11577392 / US11890758	Google "process definition graph" robot-planning family	2023–24	DAG with constraint nodes + action nodes; existence constraints restrict which children can be selected.	C2 partial (graph-constrained action selection); classical, not LLM.
LM-Nav (arXiv 2207.04429)	Shah, Osinski, Ichter, Levine;	2022	GPT-3 extracts landmarks; CLIP	C1 partial (LLM→goal); C2 partial (landmarks

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	CoRL 2022		grounds them to nodes of a topological graph built by ViNG; navigates outdoors. LLM restricted to landmarks present in the graph.	in graph). Cloud GPT-3.
SayCan / PaLM-SayCan (arXiv 2204.01691)	Ahn et al., Google	2022	LLM scores which of an enumerated closed skill set (551 primitives incl. "navigate to hard-coded location") is useful; affordance value grounds feasibility. 84% plan / 74% execution success.	C1 partial; C2 strong-conceptual (closed enumerated option set); C4 partial.
NLMap (arXiv 2209.09874)	Chen, Xia, Ichter et al., Google	2022	Frees SayCan from "a fixed list of objects nor executable options"; confirms "Vanilla SayCan provides a list of skills associated with either 1) navigation policies to hard-coded locations 2) manipulation policies... specified by object names."	Counter-teaching for C2 (treats the fixed set as a defect to overcome).
SayPlan (arXiv 2307.06135)	Rana et al., QUT; CoRL 2023	2023	LLM does semantic search over a hierarchical 3D scene graph; classical path planner + iterative replanning validated against a scene-graph simulator.	C2 partial (graph nodes); C4 partial (replanning/validation).
VLMaps (arXiv 2210.05714)	Huang, Mees, Zeng, Burgard; ICRA 2023	2023	Fuses VLM features into a 3D map; LLM translates NL to open-vocabulary spatial goals localized in the map.	C1 partial. C2 negative (open-vocabulary, not closed set).
osmAG-LLM (arXiv 2507.12753)	Xie, Schwertfeger et al. (submitted 17 Jul 2025)	2025	Zero-shot open-vocabulary object navigation; supplies the LLM "a simplified semantic-osmAG	C2 strong (structural select-from-named-nodes, no coordinates) — but not safety-framed.

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			containing only semantic nodes and hierarchical relationships (excluding coordinates and polygons)" and constrains JSON output to "max 3 nodes per room and max 3 rooms total"; navigates via ROS move_base.	
Intelligent LiDAR Nav / osmAG copilot (arXiv 2409.08493)	Xie, Schwertfeger	2024	LLM as "copilot" over an osmAG textual map + ROS move_base; PassageCostEvaluator picks a named destination.	C2 partial; integrates with ROS (move_base, not Nav2 BT).
SPINE (arXiv 2410.03035)	Ravichandran, Murali, Tzes, Pappas, Kumar; ICRA 2025	2024	Receding-horizon planner: "The planner uses an LLM to reason about subtasks... Tasks are automatically validated for safety and refined online with new map observations"; notes LLMs "may hallucinate these details."	C2 strong safety-framing (validator prevents hallucinated goals) — but allows adding new nodes/coordinates.
CANVAS (arXiv 2410.01273)	Choi et al.; ICRA 2025	2024	Converts continuous waypoints into 128 discrete tokens via K-means; reframes navigation as classification.	C2 partial (discretized closed output set) — geometric, not named, not safety-framed.
FASTNav (arXiv 2411.13262)	Chen, Han, Li	2024	Fine-tuned small LMs for multi-point navigation; JSON output of x,y coordinates; edge deployment.	C1 partial (small LM at edge). C2 negative (outputs free-form coordinates).
KnowNo (arXiv 2307.01928)	Ren et al., Princeton/Google DeepMind; CoRL 2023	2023	Conformal prediction over a calibration dataset gives a statistical task-completion	C4 strong (calibrated confidence gating of an LLM planner).

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			guarantee; if the prediction set is not a singleton, ask a human. Addresses "confidently hallucinated predictions."	
Robust Mobile Robot Path Planning via LLM (arXiv 2501.15901)	2025	llama3.1 generates/adjusts navigation waypoints dynamically, collision-free.	C1 partial. C2 negative (free-form waypoints).	
BTGenBot (arXiv 2403.12761; IROS 2024)	Izzo, Bardaro, Matteucci (POLIMI)	2024	LoRA-fine-tuned Llama2-7b/CodeLlama-7b generate BT XML on a Jetson AGX Orin; "MoveTo" node maps a location name to coordinates via a YAML file and sends the goal to Nav2.	C3 partial (LLM + BT + Nav2), but LLM is external/generative, not a BT plugin.
Interpretable Robot Control via Structured BTs + LLMs (arXiv 2508.09621)	2025	gpt-4o (via ROSA) interprets commands and selects BT behaviors; logs active BT nodes.	C3 partial (LLM + BT); cloud LLM, not a pluginlib inference node.	
Plug-and-Play LLM Knowledge Extraction (DOI 10.3390/make8020049)	Machine Learning & Knowledge Extraction, 2026	On-device DistilGPT-2 on a Jetson AGX Orin augments an FSM with asynchronous velocity suggestions gated by deadline/schema/kinematic validation; deterministic 50 Hz loop with <5 ms fallback.	C1 strong (on-device LLM, no cloud); C4 partial (validation gate).	
Das, Hussain & Nawaz (DOI 10.3390/s26020608, Sensors 2026)	"Latency-Aware Benchmarking of LLMs for NL Robot Navigation in ROS 2"	2026	Benchmarks GPT-3.5/4/5, Claude 3.7, Gemini 2.5, Mistral-7B, DeepSeek-R1, LLaMA-3.3-70B integrated with ROS 2 Nav2 across DWB/TEB/RPP; measures end-to-end latency, parse accuracy, path quality; TurtleBot4/Gazebo Fortress.	C1 partial (LLM+Nav2); C3 partial (Nav2 integration, not a BT plugin).
LocalNav (arXiv 2606.27871)	2026	Distills a frontier VLM into a 4B on-device VLM for object-goal navigation on	C1 strong (on-device).	

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			an embedded GPU (Jetson Orin).	
SafeDec (arXiv 2509.01728)	2025/26	Constrained decoding for transformer navigation policies enforcing Signal Temporal Logic safety specs at runtime without retraining.	C2 partial (constrained decoding + navigation + safety) — STL formulas, not a named-node set.	

Claim-by-Claim Analysis

Claim 1 — LLM running fully on-device (no API) resolving NL commands to navigation goals. Score: Anticipated (as a general feature). Plug-and-Play LLM Knowledge Extraction (DOI 10.3390/make8020049) runs DistilGPT-2 fully on-device on a Jetson AGX Orin; LocalNav (arXiv 2606.27871) and Qwen-RobotNav both run on-device on Jetson hardware; FASTNav (arXiv 2411.13262) targets edge deployment of small LMs for multi-point navigation. Strongest reference: DOI 10.3390/make8020049. The claim does not survive as an independent novelty; keep it as a limitation feeding the technical-effect argument (privacy, connectivity-independence, bounded latency).

Claim 2 — Closed-set annotation graph as a safety boundary; LLM output constrained to named locations only, no free-form coordinates. Score: Novel (narrowly), partially anticipated in pieces. SayCan (arXiv 2204.01691) established constraining an LLM to an enumerated option set (551 primitives); osmAG-LLM (arXiv 2507.12753) already gives the LLM only node names (excluding coordinates and polygons) and takes back enumerated named nodes in JSON; SPINE (arXiv 2410.03035) uses a validator that automatically validates tasks "for safety" and notes LLMs "may hallucinate these details." However, **no reference combines (a) a structurally closed named-node set enforced at decode/parse time, (b) an explicit anti-hallucination safety-boundary rationale, and (c) prohibition of free-form coordinate generation for a mobile-robot navigation goal.** NLMap (arXiv 2209.09874) actively teaches away — it treats the fixed set as a limitation to overcome ("without a fixed list of objects nor executable options") — which *strengthens* the inventive-step story for deliberately re-imposing a closed set for safety. Strongest references: osmAG-LLM (2507.12753) and SPINE (2410.03035). This is the anchor of your defensible gap.

Claim 3 — LLM integrated as a native Nav2 BehaviorTree plugin via pluginlib (not an external server/wrapper). Score: Novel (narrowly), obviousness risk. BTGenBot (arXiv 2403.12761) and arXiv 2508.09621 both couple an LLM to BTs and Nav2, but in each case the LLM runs *outside* the tree — as an XML generator (BTGenBot) or an external reasoning service (ROSA/gpt-4o). Das et al. (10.3390/s26020608) integrate LLMs with Nav2 but not as a BT node. **No reference places the LLM inference call inside a pluginlib-loaded nav2_behavior_tree action node.** This is genuinely undisclosed, but a skilled ROS 2 developer following the standard "Writing a New Behavior Tree Plugin" tutorial could plausibly arrive at it; on its own it risks an obviousness / Section 3(k) "mere implementation" objection. Its value derives from combination with Claim 2.

Claim 4 — Confidence calibration model trained on real robot deployment logs to correct LLM overconfidence. Score: Partially anticipated. KnowNo (arXiv 2307.01928) is squarely on point conceptually: it builds a calibration dataset from scenarios the robot encounters and uses conformal prediction to gate act-vs-ask-for-help, explicitly targeting "confidently hallucinated predictions." Your distinguishing feature is a *supervised calibration model trained specifically on the robot's own deployment logs* (versus KnowNo's out-of-the-box conformal procedure "without model-finetuning"). That specificity can survive as a dependent limitation but is not a strong independent novelty. Strongest reference: KnowNo (2307.01928).

Claim 5 — Runtime-updatable annotation graph via ROS service call without node restart. Score: Anticipated / routine. ROS *dynamic_reconfigure* and ROS 2's built-in reconfigurable parameters exist precisely to change node parameters at runtime "without having to restart the node," and online semantic-map updating is demonstrated in multiple ROS systems. Keep as a dependent claim only.

Defensible Novelty Gap (tightest non-anticipated combination)

A mobile-robot navigation apparatus in which **all** of the following hold simultaneously:

1. a language model executes **entirely on the robot's onboard compute with no network/API call;**

2. the language model's goal output is **structurally constrained at decode/parse time to a closed, enumerated set of human-named annotation nodes** of a stored annotation graph, such that generation of any coordinate or location not present in the graph is impossible by construction (the closed set functioning as an explicit safety boundary against hallucinated goals);
3. the language-model resolution is performed **within a BehaviorTree action-node plugin loaded via pluginlib into the Nav2 BT navigator** (in-tree, not as an external server), so that a resolved node is emitted directly as a **NavigateToPose / NavigateThroughPoses** goal on the blackboard; and
4. emission of the goal is **gated by a calibration model trained on the robot's own deployment logs**, which rescales the LLM's confidence and blocks/deflects (to a clarification or fallback behavior) any resolution below a calibrated threshold.

No single reference — and no obvious two-reference combination — discloses (2)+(3)+(4) together. That triad is the patentable core. (1) is included to support the technical-effect / Section 3(k) argument, not as the point of novelty.

Revised Independent Claims (patent-style)

Apparatus claim (Claim 1). An autonomous mobile robot comprising: a drive base and one or more environment sensors; an onboard processor and non-transitory memory storing an annotation graph comprising a plurality of nodes each associated with a human-readable name and a navigable pose; a language model stored in said memory and executable **solely by the onboard processor without any network or application-programming-interface call to an external computing resource**; a Navigation2 (Nav2) behavior-tree navigator; and a behavior-tree action-node plugin dynamically loaded via a plugin-class loader (pluginlib) into said behavior-tree navigator and configured to: (i) receive a free-form natural-language command; (ii) invoke the onboard language model with a decoding constraint that **restricts the language-model output to the closed, enumerated set of node names present in the annotation graph, whereby the language model is structurally prevented from generating any pose or location absent from said set**; (iii) obtain from a calibration model a corrected confidence value for the resolved node, said calibration model having been trained on logged outcomes of prior navigation attempts by the robot; and (iv) responsive to the corrected confidence value satisfying a calibrated threshold, write the navigable pose of the resolved node to the behavior-tree blackboard as a navigation goal for execution by the Nav2 stack, and responsive to the corrected confidence value failing said threshold, trigger a fallback or clarification behavior node instead of issuing a navigation goal.

Method claim (Claim 2). A computer-implemented method for constraining natural-language robot navigation, comprising: storing, on a mobile robot, an annotation graph of named nodes each mapped to a navigable pose; receiving a free-form natural-language navigation command; resolving the command to a target node by executing a language model **entirely on the robot's onboard processor without any external network call**, wherein the resolving applies a constraint that limits the language model's admissible output to the closed enumerated set of node names in the annotation graph such that no coordinate or location outside said set can be produced; computing a calibrated confidence for the resolved target node using a calibration model trained on the robot's own logged navigation outcomes; and, only when the calibrated confidence satisfies a predetermined threshold, emitting the navigable pose of the resolved node as a Nav2 navigation goal from within a behavior-tree action-node plugin loaded via pluginlib into the robot's behavior-tree navigator; otherwise invoking a clarification or fallback behavior; wherein the resolving, the confidence computation and the goal emission all occur within the behavior-tree execution tick.

Dependent Claims

- **D1.** The apparatus of Claim 1, wherein the decoding constraint is enforced by grammar-constrained or JSON-schema-constrained decoding that admits only tokens forming a node name in said closed set.
- **D2.** The apparatus of Claim 1, wherein the annotation graph is updatable at runtime via a ROS 2 service call that adds, removes, or renames nodes without restarting the behavior-tree navigator node.
- **D3.** The apparatus of Claim 1, wherein the calibration model comprises a post-hoc rescaling (temperature/Platt-style) mapping from raw language-model token or verbalized confidence to empirical success probability learned from the robot's deployment logs.
- **D4.** The apparatus of Claim 1, wherein failing the calibrated threshold causes the plugin to return a prediction set of two or more candidate named nodes and to solicit a human disambiguation selection.
- **D5.** The apparatus of Claim 1, wherein the onboard language model is a quantized model of fewer than eight billion parameters executing on an embedded GPU or CPU of the robot, and wherein end-to-end resolution latency within the behavior-tree tick is bounded by a configured timeout after which a deterministic fallback node executes.

- **D6.** The apparatus of Claim 1, wherein the annotation-graph nodes further encode hierarchical parent relationships (rooms within floors) and the decoding constraint is scoped to the subgraph reachable from the robot's current node.
- **D7.** The method of Claim 2, further comprising logging, for each resolved goal, the command, the resolved node, the calibrated confidence, and the achieved outcome, and periodically retraining the calibration model on the accumulated logs.
- **D8.** The apparatus of Claim 1, wherein the behavior-tree action-node plugin exposes the resolved node and its calibrated confidence as behavior-tree output ports consumable by downstream condition and control nodes of the same tree.

Freedom-to-Operate (FTO) Assessment

- **Google's SayCan patent family (US20230311335A1 / WO2023192497A1)** is the most important to clear. Its claims recite selecting a robotic *skill* using a task-grounding measure (skill-description probability in the LLM output) combined with a world-grounding (affordance) measure. If your product resolves a *named navigation node* (not a skill) via a closed-set decode constraint and a *calibration* model (not an affordance value function), you likely do not read on those claims — but "navigate to X" appears as an enumerated skill in their embodiments, so the granted claim language must be examined in full. **Action: obtain the granted/allowed claim set and run a claim chart before commercial launch.**
- **US20210086353A1 / US11972339B2 (free-form NL → waypoints, Google)** uses deep-RL, with no LLM and no closed graph; low blocking risk for an LLM+closed-graph product, but monitor continuations.
- **Locus Robotics US9758305B2 / US10019015 (semantic mapping SKU→fiducial→coordinates)** is warehouse-specific and language-model-free; low risk unless you build a warehouse SKU-picking product.
- **US11813751 / US11577392 / US11890758 (Google process-definition-graph planning)** cover graph-constrained action selection but classically; low risk for the LLM-resolution path, moderate only if your graph-traversal method is itself claimed broadly.
- **Constrained-decoding infrastructure** (Outlines/XGrammar and OpenAI/Google/Anthropic structured-output features) is largely method/service tooling; using such libraries is unlikely to infringe a robot-navigation *apparatus* claim, but verify no asserted method patents on grammar-constrained decoding reach downstream use.
- **Overall FTO posture:** No identified patent squarely blocks the (closed-set + in-tree LLM + calibration) combination. The principal residual risk is the SayCan family; mitigate by claim-charting and by drafting your claims around "named navigation-node resolution" + "calibration model" rather than "skill affordance."

Filing Strategy

- **Route:** File an **Indian provisional at the Indian Patent Office first** to secure the earliest priority date at low cost, then file a **PCT within 12 months** (the IPO can act as Receiving Office / ISA / IPEA). Enter national phase (US, EP, CN, plus any target markets) by **31 months** from priority. Note that under the Patents (Amendment) Rules 2024 (effective 15 March 2024), Indian national-phase entry and the Request for Examination (Form 18) both fall at 31 months and must be filed *together*; missing that deadline treats the application as withdrawn, with restoration generally unavailable.
- **Section 3(k) survival:** India's 2025 CRI Guidelines (issued March 2025, finalized July 2025) require a demonstrable **technical effect** beyond a "computer program / algorithm per se," applying a single structured technical-effect test. Anchor the specification and claims on concrete technical effects: (i) elimination of hallucinated / out-of-graph goals as a *physical safety* guarantee for a moving robot; (ii) bounded, deterministic in-tick latency from on-device inference feeding a real-time control loop; (iii) improved goal-success reliability from deployment-log calibration. Recite the hardware (drive base, sensors, embedded processor) in the independent apparatus claim.
- **Recommended classifications:** primary **G05D1/02** (control of position/course of land vehicles/robots) and current subgroups; **B25J9/1602 / B25J9/163 / B25J9/1697** (program-controlled manipulators; learning/adaptive control; vision-controlled systems); **G06F40/30 / G06F40/40** (semantic analysis / NL processing); **G06N3/04 / G06N3/08** (neural-network architecture / learning); **G10L15/18 / G10L15/22** if voice input is claimed; **G05B2219/40** robotics program-control subgroups.
- **Fee posture:** If the applicant is a DPIIT-recognized Indian startup or a small entity, claim the substantial official-fee concessions.
- **Estimated timeline:** Provisional → complete specification within 12 months; PCT filing at ~month 12; International Search Report + Written Opinion by ~month 16–18; national-phase entries by month 31; first substantive examination reports typically 12–36 months after the examination request depending on jurisdiction (Indian examination can be expedited if the applicant qualifies, e.g., startup/DPIIT). Realistic first grant in the fastest jurisdiction ~3–4 years from priority.

Recommendations

1. **File the Indian provisional now** around the (closed-set-as-safety + in-tree Nav2 BT plugin + deployment-log calibration) triad — this is the only defensible core. Do **not** lead with "on-device LLM" or "runtime-updatable map" as independent novelties; both are anticipated (DOI 10.3390/make8020049; ROS 2 reconfigurable parameters).
2. **Before launch, obtain and claim-chart the granted claims of US20230311335A1 / WO2023192497A1** (SayCan family). If any independent claim reads on "resolving a navigation goal," design around by emphasizing named-node resolution + calibration rather than skill-affordance selection.
3. **Draft the specification with explicit technical-effect language** (physical safety, bounded latency, reliability) to clear Section 3(k) and the EPO COMVIK inventive-step test used for parallel European prosecution.
4. **Generate and preserve real deployment logs** and describe the calibration-model training pipeline in enabling detail — this is both the distinguishing feature over KnowNo (which is explicitly "without model-finetuning") and a Section 3(k) technical-effect anchor.
5. **Benchmarks/thresholds that would change the strategy:** if a full-text patent-database search (USPTO / Espacenet / PATENTSCOPE on G05D1 + G06N) surfaces a filing reciting closed-set LLM goal restriction *for safety*, narrow to the calibration-gated in-tree-plugin combination; if the osmAG-LLM or SPINE authors file a patent claiming named-node restriction, accelerate your priority date and make the calibration + in-tree limitations the explicit point of novelty.

Caveats

- **Web-based patent coverage is incomplete.** A formal full-text search of USPTO Patent Center, Espacenet, and WIPO PATENTSCOPE on the relevant CPC classes (G05D1, G06N, B25J9/16, G06F40) should be run before relying on "no blocking patent found." Recently filed but unpublished applications (18-month secrecy period) cannot be ruled out.
- **Several 2025–2026 arXiv references are very recent** (osmAG-LLM 2507.12753, SafeDec 2509.01728, LocalNav 2606.27871, Das et al. 10.3390/s26020608). Confirm each publication date against your intended priority date — anything published before your filing date is citable prior art against you, so an early provisional is time-critical.
- **Claim 3 (in-tree BT plugin) carries obviousness risk** given Nav2's documented plugin-authoring workflow; its patentable weight comes almost entirely from combination with Claim 2.
- **The SafeDec (arXiv 2509.01728) and CANVAS (arXiv 2410.01273) references show the "constrained decoding for navigation" concept is actively developing;** treat the closed-set-*for-named-nodes-as-safety* framing as the narrow inventive distinction, not constrained decoding in the abstract.
- **This is a technical prior-art and strategy assessment, not legal advice.** Engage a registered Indian patent agent/attorney for claim finalization, a professional novelty search, and a formal FTO opinion before filing or launch.